

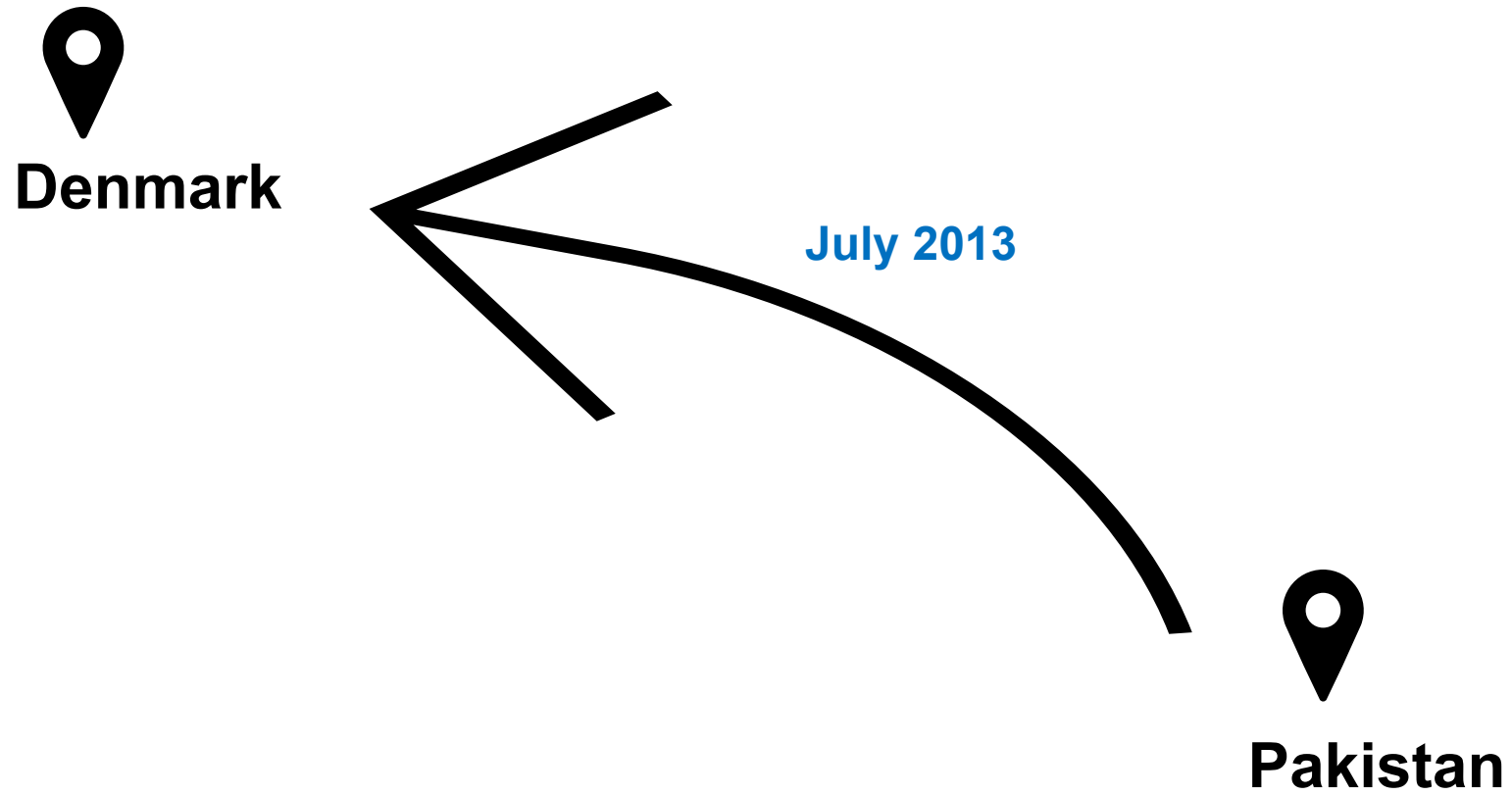
SCIENTIFIC METHODS

AISHA UMAIR

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ABOUT ME

AISHA UMAIR



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- PhD. in Software Engineering from Syddansk Universitet (SDU)
PhD Project: Coordination Protocol for System of Cyber-Physical Systems
- Associate Professor and Vice Head of Education, SDU Software Engineering, MMMI

Research Interests:

- Multi-Objective Optimization
- Coordination & Negotiation Protocols
- Agent Based Systems
- Scalable Teaching

Responsibilities

➤ Teaching

- Scientific Methods (Master: 1st Semester)
- Advanced Object-Oriented Programming (Bachelor: 2nd Semester)

➤ Supervision

- Internship Projects' supervision

➤ Administrative Tasks

- BS Semester-1 SE/ST Coordination
- Academic assessment of credit transfers for study-start, semester exchange, master's application, approval of InCompany projects etc.....

➤ Educational Research Projects

- Automated feedback and corrections
- VSLab - Virtual Software Laboratories as a Scalable Research Infrastructure for Higher IT Educations
- Scalable Exams at SDU for Digital Skills

MAIN CONTACT

- Aisha Umair (aiu@mmmi.sdu.dk)
- If you have any queries outside the lecture, please send an e-mail.
- If face-to-face meeting is required outside the lecture, please book an appointment

Teaching Assistants (TAs)

Responsible for exercise hours and providing feedback on your exercise solutions.

- Toke Odgaard-Jans toodg22@student.sdu.dk
- Mohammed Mansoor Ahmed moahm25@student.sdu.dk

Agenda

- ✓ Course Information
- ✓ Course Format
- ✓ Mini Research Project
- ✓ Mandatory Deliverables
- ✓ Final Exam
- ✓ Groups formation
- ✓ Use of GenAI
- ✓ Course Evaluation
- ✓ Research Topics

COURSE INFORMATION

LEARNING OBJECTIVES

The student is able to:

- Independently **find literature** on a specific research topic and conduct systematic literature review
- Identify **qualitative and quantitative research** approach and choose between them appropriately
- Explain **qualitative data collection and data analysis** approaches
- Explain **quantitative data collection and data analysis** approaches
- Demonstrate **ethical approach** and research integrity in their own research
- Identify **common threats** to quality of research and describe the validity and reliability of their own research
- Present, **both orally and in writing**, the research process for a specific research topic, findings, and conclusion in a coherent and concise manner.
- Write a **scientific report** covering a specific research topic using academic writing format and required content.

TOPICS TO BE COVERED

- ▶ Introduction to research approaches
 - ▶ Qualitative and Quantitative
- ▶ Systematic Literature Reviews
- ▶ Qualitative and Quantitative Data Collection and Data Analysis Methods
- ▶ Research Ethics
- ▶ Validity Threats
- ▶ Tips on Academic Writing

WHAT YOU WOULD BE DOING?

- Attend lectures, participate in in-class tasks/discussions, complete mandatory assignments
- You will do a mini research project
 - Choose a research topic
 - Find (relevant) literature for the given topic
 - Plan and design the study
 - Execute the study
 - Present the results of your study both oral and written
 - Perform peer-evaluation on each others' work

COURSE FORMAT

- 12 lectures
- 10 lectures (2h lecture + 2h class activity)
- Class activity could be mandatory assignments, in-class tasks or project work
- Class activities:
 - 4 mandatory assignments (submission required)
 - 3 in-class tasks (no submission required)
 - Group project presentation & peer feedback

Note: The last two lectures are reserved for project presentations

SCHEDULE

- W. 36 (3rd Sept) – Welcome and Introduction to the course
- W. 37 (10th Sept) - Introduction to research and philosophical views (**Assignment 1**)
- W. 38 (17th Sept) - Theory, research problem, and selecting research approach (**Assignment 2**)
- W. 39 (24th Sept) - Evidence based software engineering (In Class Task)
- W. 40 (1st Oct) - Guest lecture on information retrieval (**Submit: research plan on 5th Oct**)
- W. 41 (8th Oct) – Research ethics (In Class Task)
- W. 42 (15th Oct) - (No lecture)
- W. 43 (22nd Oct) - Qualitative research methods
- W. 44 (29th Oct) - Quantitative research methods (**Assignment 3**)
- W. 45 (5th Nov) - Validity threats (In Class Task)
- W. 46 (12th Nov) - Peer review and academic writing
- W. 47 (19th Nov) – No Lecture
- W. 48 (26th Nov) – No Lecture (**Submit: report draft: 25th Nov**)
- W. 49 (2nd Dec) – Project presentations
- W. 49 (3rd Dec) – Project presentations
- W. 51 (18th Dec) - **Submit final report (Assignment 4)**

Mini Research Project

You need to plan, execute and report a research project.

Research plan.

- You need to write a plan that describe the topic,
- the study aim,
- research question,
- what method you choose,
- what you expect to produce and
- the time plan.

Project Presentation:

- You also need to present your project verbally and be prepared for a Q&A.
- When it's not your day of presentation, you will act as an evaluator of another group's project.

Research report

You need to write a report of the methods and results of your research project.

PEER EVALUATION

- ▶ You will evaluate a report from another group
 - ▶ Which group to evaluate? (will be decided later)
 - ▶ Come up with suggestions for improvements for their report.
 - ▶ Also, think about questions that you ask them to clarify which is not clear from the report.
 - ▶ Be nice and constructive.
 - ▶ Suggestions regarding spelling and grammars ONLY are not acceptable.

DELIVERABLES

- **Assignments**, 3 of the lectures include assignments (mandatory but not graded).
 - Low effort report, where you need to show that you understood the concept, completed and submitted the assignment.
- **Research Plan** (Mandatory but not graded)
- **Draft Report** (Mandatory but not graded)
- **Final Report- 4th Assignment** (Mandatory but not graded)

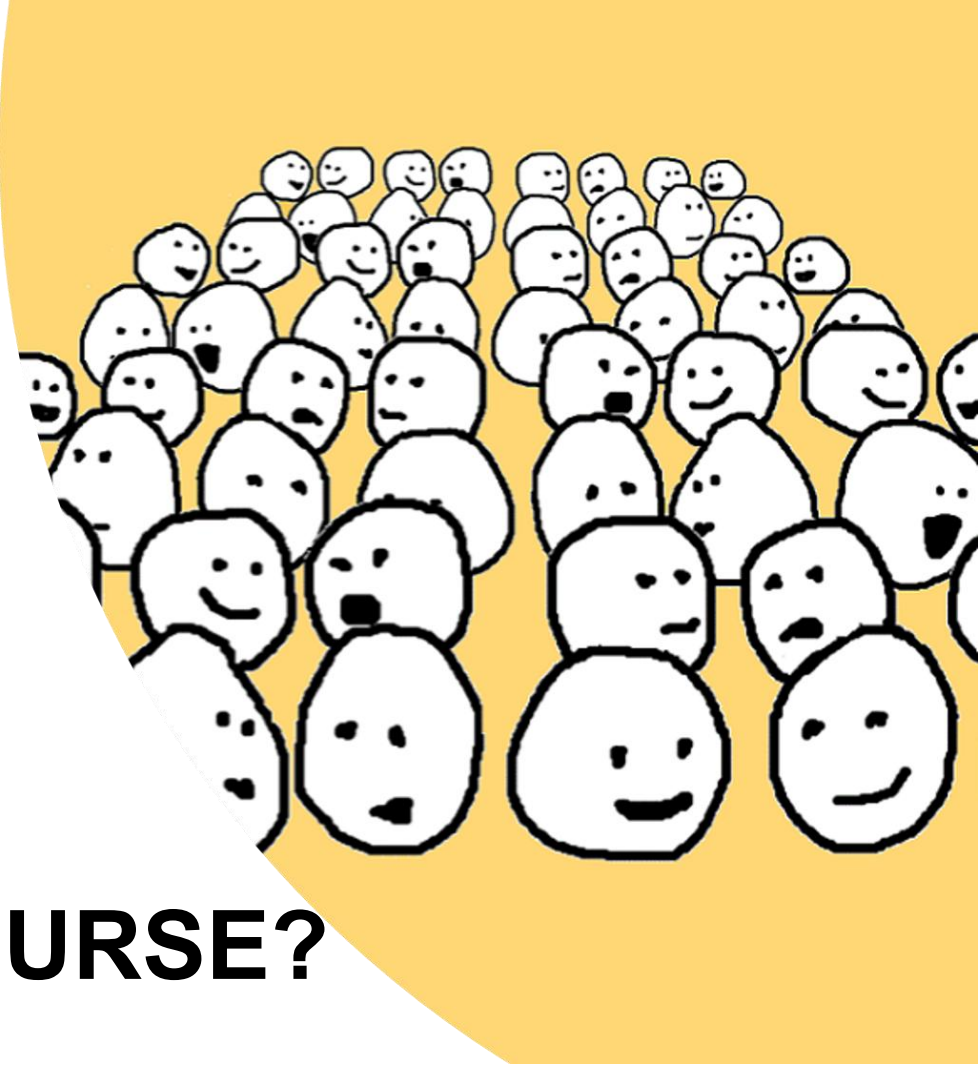
Note: All deliverables are submitted through Itslearning

Final Exam

Written Exam

- **Duration:** 2 hrs
- **Exam Aid:** All mandatory assignments (portfolio) are allowed to be used in the final exam.
- **Grading:** 7-point grading scale

Think – Talk - Share



WHAT DO I GET OUT OF THIS COURSE?

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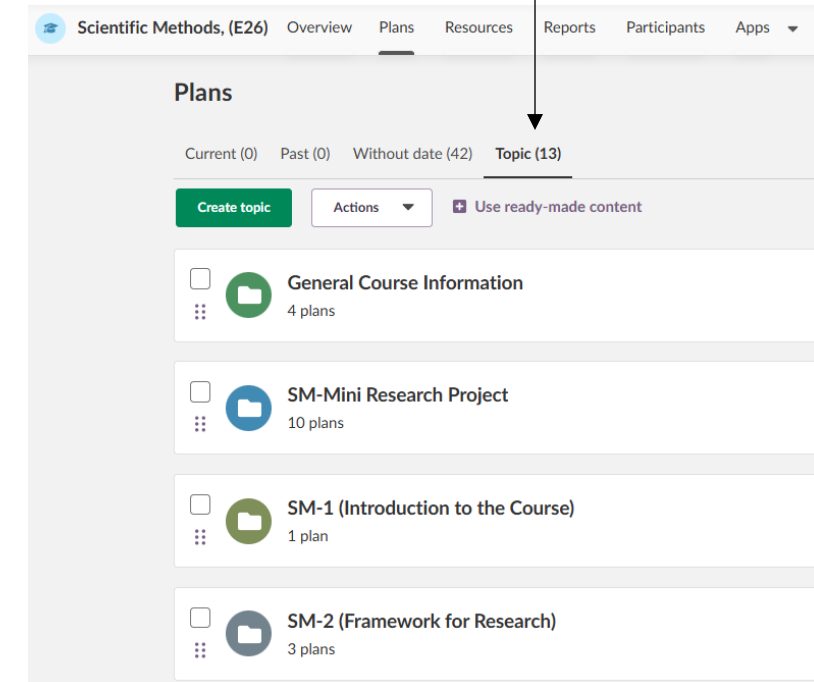
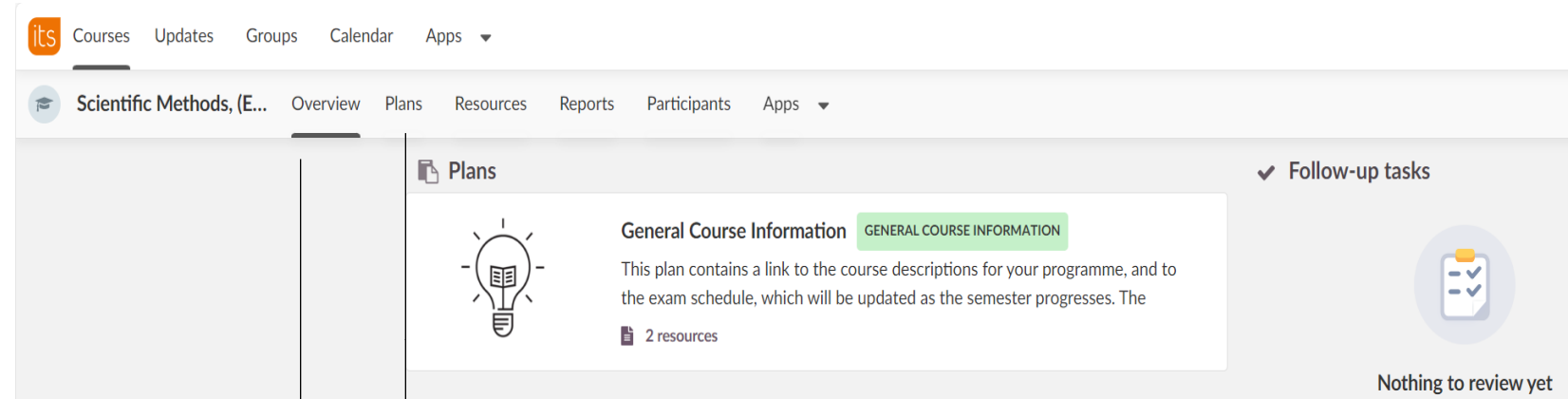
WHAT IS IT GOOD FOR?



Break (10 min)

itslearning

- Post announcements (**Overview tab**)
- Assignment description and submission (**Plans tab**)
 - Project proposal submission
 - Draft report submission
- In-Class Tasks description (**Plans tab**)
- Slides, reading materials, and links to reading materials (**Plans tab**).



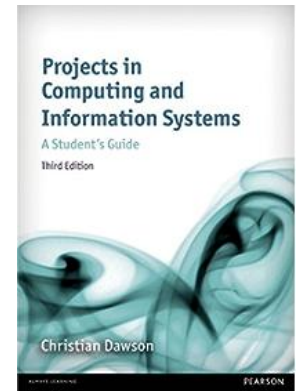
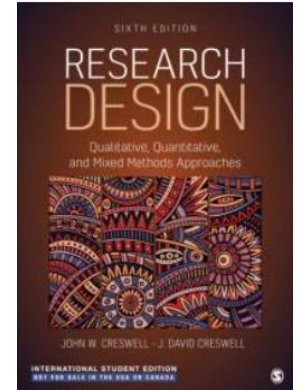
IMPORTANT DATES

- Research plan: 5-10-2026
 - Submitted via itslearning
- Research report draft: 25-11-2026
 - Submitted via itslearning + to your peer group
- Presentation: 2-12-2026 and 3-12-2026
- Research report final: 18-12-2026
 - Submitted via itslearning

LITERATURE

Main:

- Creswell J.W., and Creswell, J.D. *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, Sixth Ed. Sage Publishing
- Dawson, C. (2009) *Projects in Computing and Information Systems: A Student's Guide*, third Ed. Addison Wesley.
- Supplementary Material:
 - Will be provided on ongoing basis corresponding to the topic.



Groups Formation

4-6 members in a group (ideally 6)

- Please register yourself in groups

Go to Plans -> Topic -> Mini-Research Project -> Group Registration

Important: Group Sign-up Instructions

When you register for your project groups online, please do so sequentially.

Find the lowest-numbered empty group available and have all your members register for that same group.

For example, if Groups 1-4 are full/taken, your group should sign up for Group 5.

Please do not choose a random group number or leave gaps between filled groups. This helps me keep the group list organized.

Thank you for your cooperation 😊



Use of GenAI

NO AI (Final Exam)

The assessment is completed entirely without AI assistance in a controlled environment, ensuring that students rely solely on their existing knowledge, understanding, skills and supplementary exam aid.

You must not use AI at any point during the assessment. You must demonstrate your core skills and knowledge.

Use of GenAI

AI COLLABORATION (In-Class Tasks, Mandatory Assignments & Mini-Research Report)

AI may be used to help with idea generation, feedback, and refinement. Students should critically evaluate and modify the AI suggested outputs, demonstrating their understanding.

You may use AI to assist with specific tasks such as brainstorming, refining and evaluating your work. You must critically evaluate and modify any AI-generated content you use.

Recommended Course: Learn GenAI

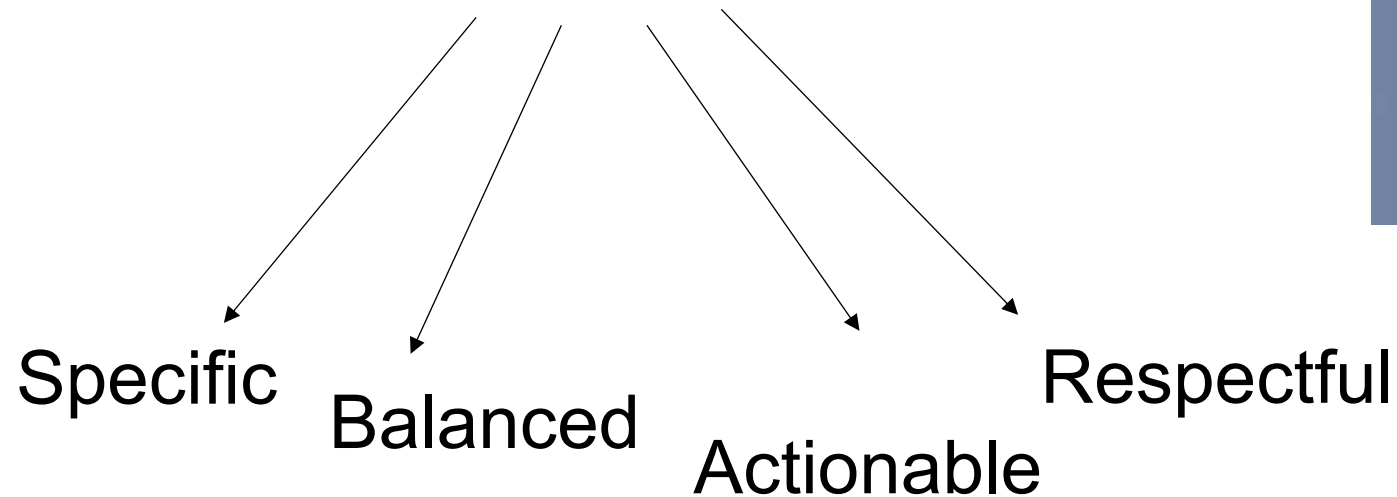
Online Self-paced Course on the use of Generative Artificial Intelligence

This course is designed by SDU, Aarhus University and Aalborg University to introduce students to the fundamentals of GenAI and demonstrate how you can leverage these powerful tools in your studies and future careers as software engineers. This course is available online at:

<https://teachgenai.au.dk/learn-genai/learn-genai-course>

Course Evaluation

- Your feedback matters.
- Kindly give **Constructive** course evaluations.



Last not but not the least 😊

Let's commit to building and maintaining a culture of mutual respect.
This applies to every interaction we have - with the teacher, teaching assistants and with fellow students.



Research Topics

- I can also suggest topics
- You can also choose topics of your interest.
- Topics???

Next Lecture Topic:

Framework for research

Any Question?????